

MODEL EVALUATION

GLY606 Water Data Analysis & Modeling

Nov 18th 2024



Final project

- **Topic:** Hydrologic modeling and evaluation: a case study
- **Oral presentation (20%)**
 - Date: Dec 6th 2024 (Friday)
 - Format: 15 min (11 min presentation + 4 min Q&A)
- **Final report (80%)**
 - Due date: Dec 11th 2024 (Wednesday)

A powerpoint-type presentation, like you would present at a conference. Your classmates are your audience. Present your question, why it is important, what methods you employed to answer the question, your major results, and the implications of your results.

- **Clarity of Overall Presentation (10 points)**
 - A clear introduction to your river basin
 - Clear explanations of what you did, easy to follow
 - Model setup, model evaluation
 - Good transitions from one section to the next
 - Ability to answer questions raised by audience
- **Quality of Visual Aids/Graphics (10 points)**
 - All graphics easy to see, properly labeled, large enough font
 - Correct limits on plot axes to show relevant information
 - Proper choice of type of graph to convey information
 - Interest level – slides nice/interesting to engage audience
- **Technical Analysis and Discussion (10 points)**
 - Proper methods applied
 - Accuracy of results
 - Explanation of why results are what they are
 - Ability to relate results to other studies and other presentations in the class

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For the written report, you must include an abstract (no more than 300 words) and a list of references; these can be in addition to your 10 (for a group of two; or 15 for a group of three) pages. Follow the same format outlined for the presentation but feel free to include more details (equations, etc.). Please address any questions or comments that were raised during the presentation. Use clear, concise sentences and proper grammar.

- **Quality of Writing (10 points)**

- Clear and easy to read
- Proper grammar and spelling
- Clear statement of the problem
- Good transitions from one section to the next
- Clarity of explanations of what you did

- **Technical Analysis (10 points)**

- Clearly described methods, proper literature cited
- Technical difficulty of methods applied
- Accuracy of results

- **Interpretation, Analysis, and Discussion of Result (10 points)**

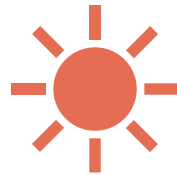
- Attempted to explain why certain biases were observed in the model results
- Thought about results in context of other studies
- Outlined next steps to be taken
- Addressed the underlying assumptions when we set up the model and how these may have affected results

How to evaluate a hydrologic model?



Overall biases

Water quantity
RMSE
Correlation of variation
MAE
Nash Sutcliffe Efficiency (NSE)
Kling-Gupta Efficiency (KGE)



Seasonality



Interannual variability

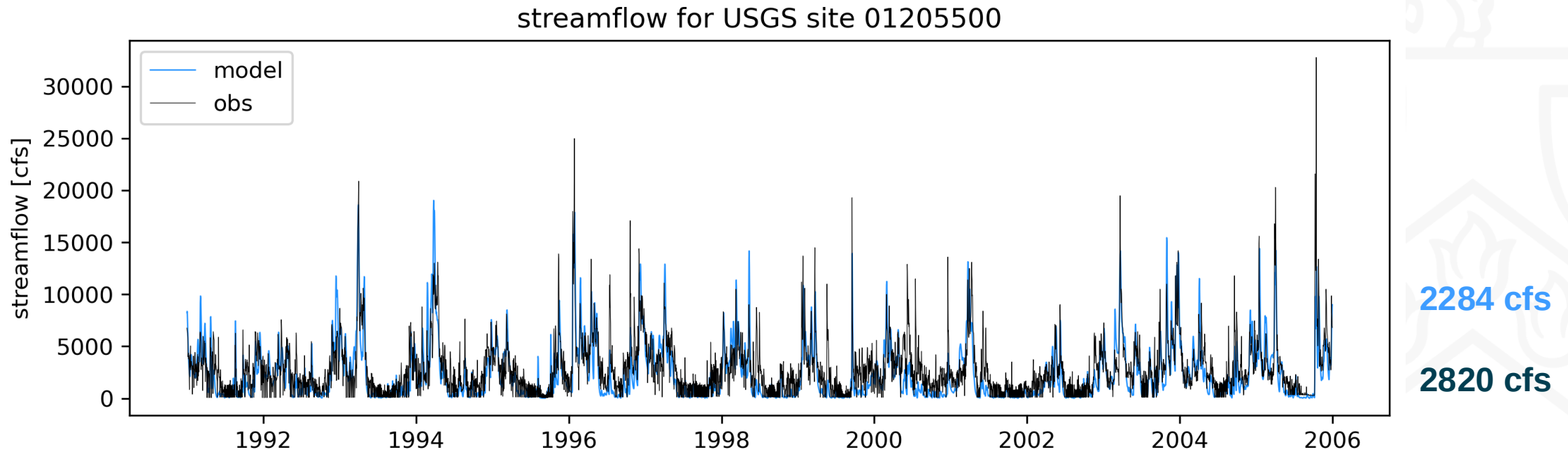


Evaluate extreme values?

Flood
Drought

Bias in water quantity

- We usually use the biases in mean annual streamflow as our evaluation of bias in water quantity (water availability)



Root Mean Square Error (RMSE)

- RMSE is a commonly used metric to measure the differences between values predicted by a model and the values observed. It gives a sense of how well the model is performing; lower values indicate better fit.

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (y_i^{model} - y_i^{obs})^2}{N}}$$

You can imagine that sites with larger flows tend to have large RMSEs.

How can we compare across multiple sites?

Correlation of Variation (CV)

- CV can be used to assess the relative variability of the model's predictions compared to the mean of the observed data. A lower CV indicates less variability relative to the mean, which generally suggests a more consistent model performance.

$$CV = \frac{RMSE}{\overline{y^{obs}}}$$



Mean Absolute Errors (MAE)

- How do we calculate MAE?

$$MAE = \sum_{i=1}^N |y_i^{model} - y_i^{obs}|$$

- The RMSE will always be larger or equal to the MAE; the greater difference between them, the greater the variance in the individual errors in the sample.

Recommended reading: Hodson, T. O. (2022). Root mean square error (RMSE) or mean absolute error (MAE): when to use them or not. GMD. [doi:10.5194/gmd-2022-64](https://doi.org/10.5194/gmd-2022-64)

Nash-Sutcliffe Efficiency (NSE)

- The NSE is used to assess the predictive skill of hydrological models

$$\text{NSE} = 1 - \frac{\sum_{t=1}^T (Q_o^t - Q_m^t)^2}{\sum_{t=1}^T (Q_o^t - \bar{Q}_o)^2}$$

where $\bar{Q}_o$ is the mean of observed discharges, and Q_m is modeled discharge. Q_o^t is observed discharge at time t .

NSE ranges from $-\infty$ to 1.

NSE	What it means?
1	Perfect model
0.5	Model simulation can be judged as satisfactory if $\text{NSE} > 0.50$ (Moriassi et al., 2007)
0	Model skills are the same as the mean value of observations

Disadvantage of NSE

- Sensitivity to Outliers:** NSE can be overly sensitive to extreme values or outliers in the dataset, which can disproportionately affect the efficiency score
- Performance in Low-Flow Conditions:** NSE may not adequately capture model performance during low-flow conditions, which can be critical in hydrological studies

Kling-Gupta Efficiency (KGE)

- The Kling–Gupta efficiency (KGE; Gupta et al., 2009) is based on a decomposition of NSE into its constitutive components (correlation, variability bias and mean bias), addresses several perceived shortcomings in NSE

$$KGE = 1 - \sqrt{(r - 1)^2 + (\alpha - 1)^2 + (\beta - 1)^2}$$

r is the linear correlation between observations and simulations, α a measure of the flow variability error, and β a bias term

$$\alpha = \frac{\sigma_{sim}}{\sigma_{obs}} \quad \beta = \frac{\mu_{sim}}{\mu_{obs}}$$

KGE ranges from $-\infty$ to 1.

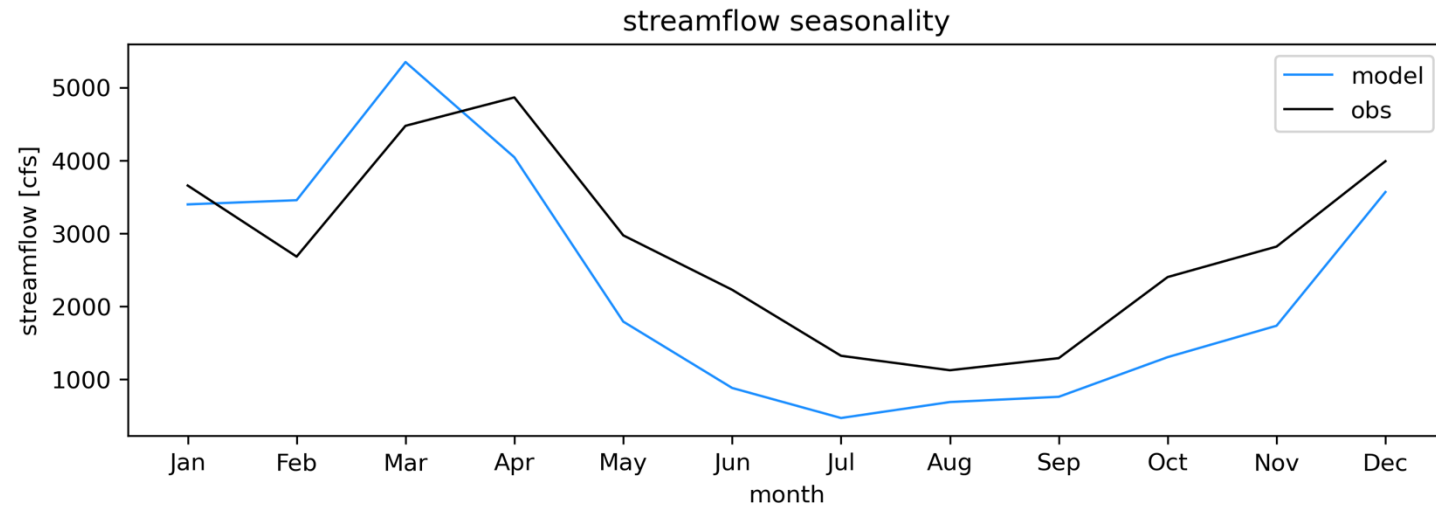
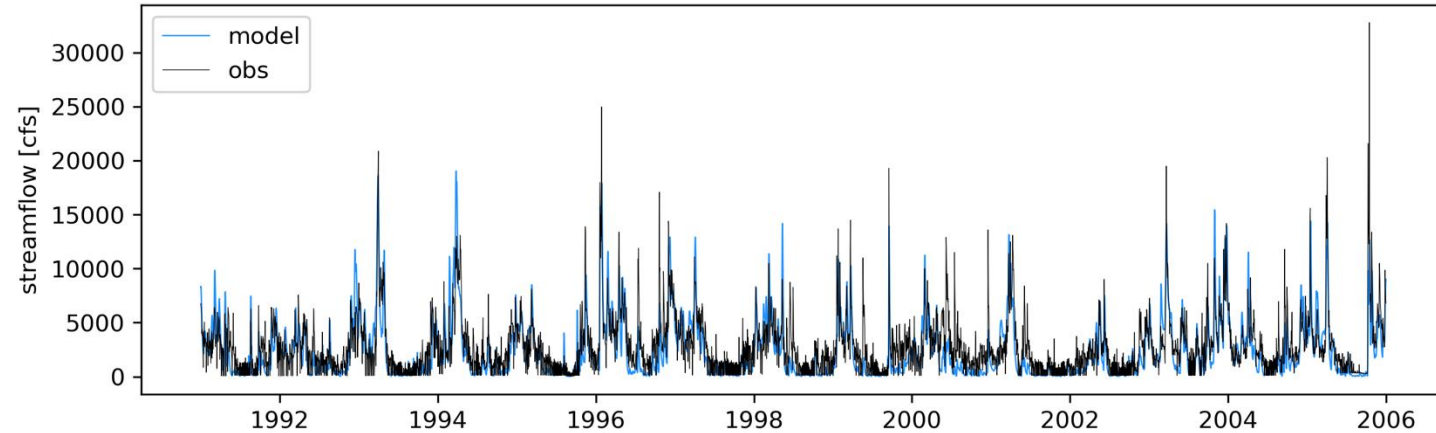
Does **KGE = 0** means the same thing as **NSE = 0**?

No! Using the mean flow as a predictor does not result in $KGE = 0$, but instead $KGE = -0.41$

Recommended reading: Knoben et al. (2019). Technical note: Inherent benchmark or not? Comparing Nash-Sutcliffe and Kling-Gupta efficiency scores. HESS. [doi:10.5194/hess-23-4323-2019](https://doi.org/10.5194/hess-23-4323-2019)

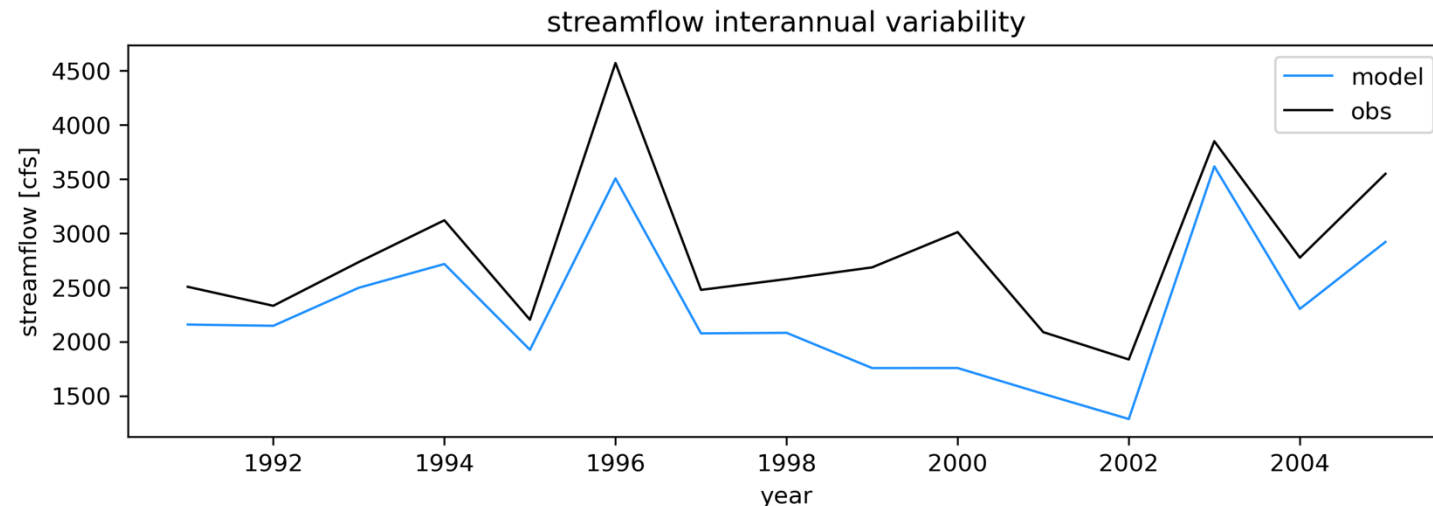
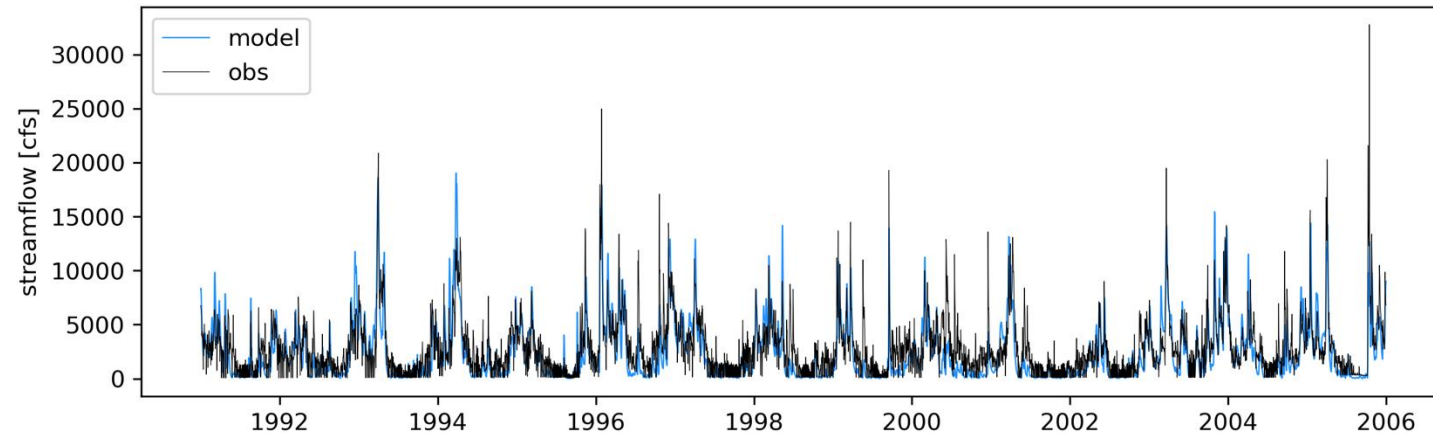
Seasonality

- Streamflow seasonality refers to the predictable patterns of streamflow variations throughout the year
 - We usually calculated the multi-year mean monthly streamflow to represent the streamflow seasonality



Interannual variability

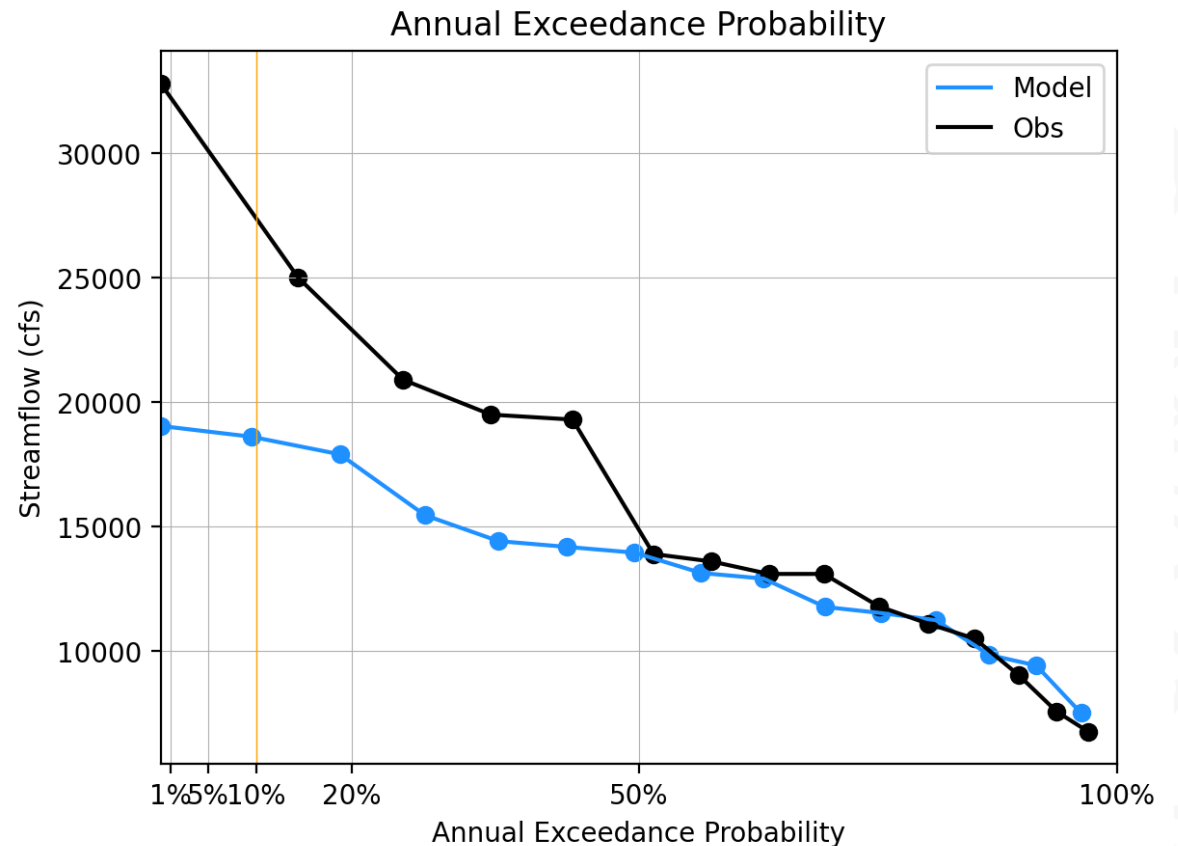
- Streamflow inter-annual variability refers to the variations in streamflow from one year to the next.
 - We usually use the time series of annual mean flow to represent the streamflow inter-annual variability
 - We used the correlation coefficient to quantify how strong the relationship is, which also evaluate the robustness of the model's predictability



Correlation Coefficient is 0.898

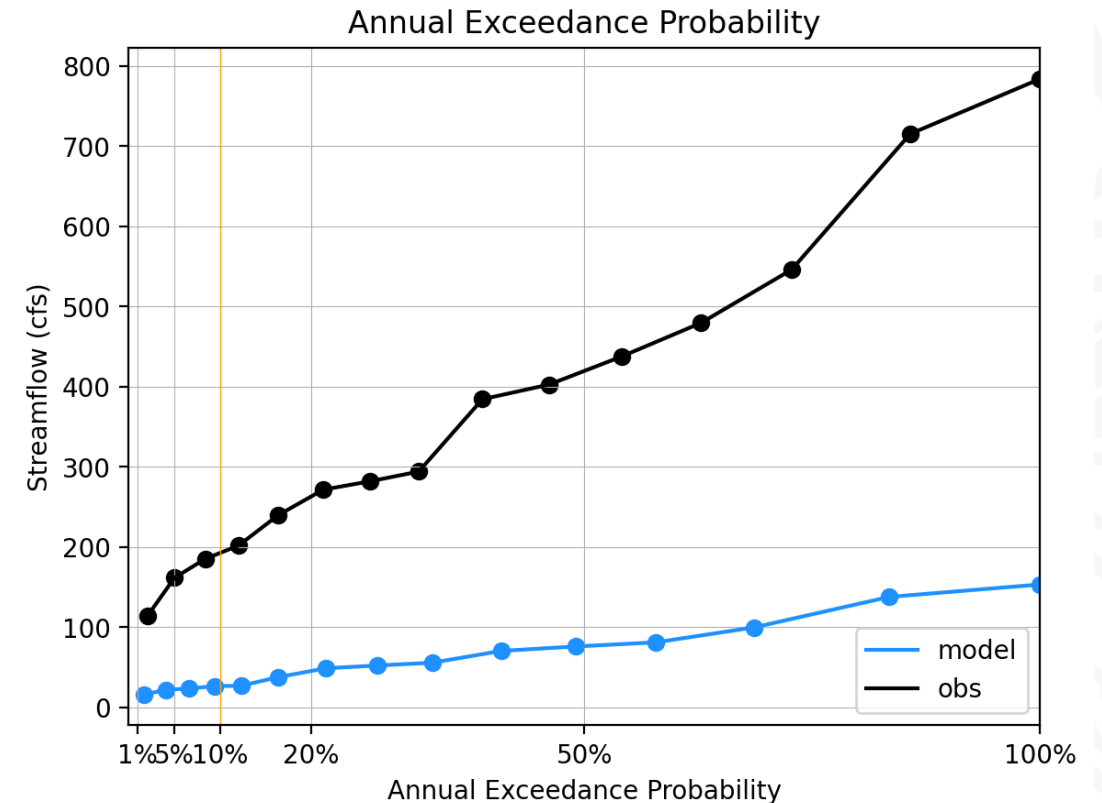
Extreme values – Flood

- 1. Risk Management:** Accurate flood predictions enable better risk management and preparedness, helping to mitigate the impacts of floods on communities and infrastructure
- 2. Early Warning Systems:** Reliable flood forecasting is crucial for early warning systems, which provide timely alerts to authorities and the public, allowing for proactive measures to reduce damage and save lives



Extreme values – Low flows

- 1. Agricultural Planning:** Farmers can use drought forecasts to plan their planting and irrigation schedules, reducing the risk of crop failure and optimizing water use
- 2. Environmental Protection:** Predicting droughts helps in protecting ecosystems that are vulnerable to water shortages, maintaining biodiversity and ecological balance



Can we summarize how well our model performs in this basin?

What are the potential sources of errors in the model?